

CCNCS Summer Internship 2025

Final Review

Project Title : Hybrid Privacy Approach for Federated Learning in Healthcare using Anonymization and Homomorphic Encryption

Project ID : 22

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Outline

- Abstract and Scope of the Project
- Suggestions from Review - 2
- Implementation Details
- Project Demonstration
- Test Plan and Strategy
- Results and Discussion
- Project Report Draft
- Conclusion and Future Work
- References

Abstract and Scope

- The growing use of mental health survey records in machine learning raises serious concerns about patient privacy, even in distributed settings like Federated Learning. Traditional anonymization alone may be insufficient against inference attacks, while homomorphic encryption (HE), though powerful, is computationally intensive and underutilized in real-world collaborative healthcare applications. This research aims to design a framework that combines anonymization and HE to ensure robust privacy protection during distributed model training using Federated Learning on sensitive medical data, enabling secure learning without exposing identifiable or intermediate information.
- The project explores a hybrid privacy-preserving framework that combines data anonymization techniques (k-anonymity, l-diversity) with homomorphic encryption (HE) in a federated learning setup for mental health survey records, to protect gradient updates during communication. The architecture is client-server, using data. The output model performs risk stratification of patients.

Suggestions from Review - 2

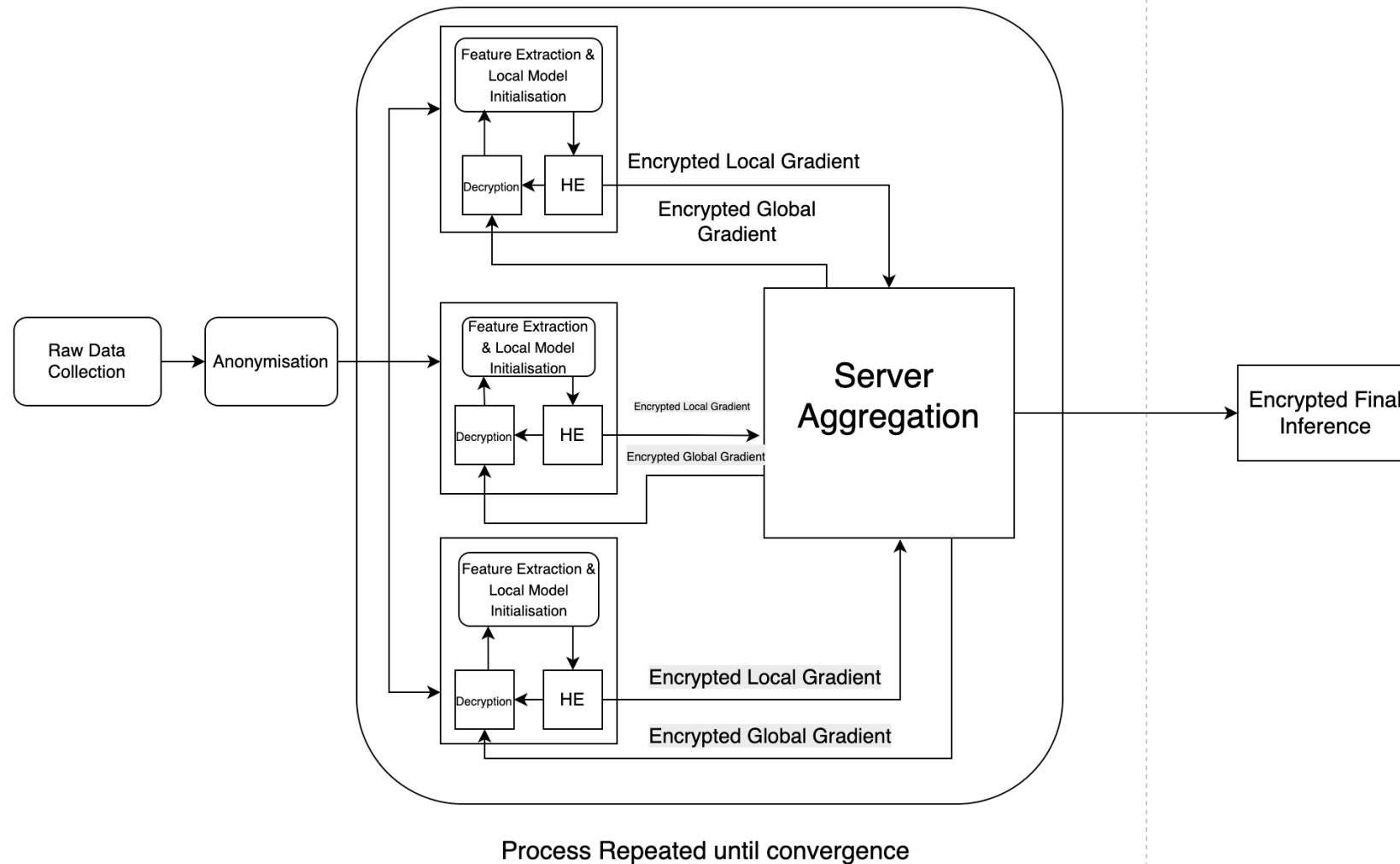
Suggestions:

- Correct the flaw in the pseudocode to reflect the correct logic - The decryption step was mistakenly placed in the server logic, which violated privacy principles.
- Identify Novelty / Uniqueness in Proposed Methodology - Recommended us to highlight the novelty of our framework, especially any new variations or tuning done in anonymization/encryption techniques.

Feasibility and Progress on the Suggestions:

- Pseudocode corrected by shifting decryption key to the client side as per the feedback.
- Novelty identification in progress through DP parameter tuning and looking into the other parameters which could make our framework unique.

Model Architecture Workflow



Implementation Details

Module 1: Data Collection & Preprocessing

Data Collection

- **Dataset:** *Depression, Anxiety and Stress Scales (DASS)*
- **Source:** OpenPsychometrics.org (open-source psychological survey data)
- **Format:** CSV with Likert scale responses (1–4) to 42 questions
- **Simulation:** Split into 3 parts to represent 3 hospitals for federated setup

Preprocessing Steps

- Removed incomplete entries (missing answers)
- Converted categorical responses to numerical format
- Normalized scores to range [0, 1]
- Generated derived fields (e.g., Condition from raw scores)

Module 2: Data Anonymization

Tools used : Pandas , numpy, ARX , hashlib

Process

- Removed direct identifiers (if any)
- **Selected quasi-identifiers** (e.g., age, gender, region) for anonymization
- **Generalized** the age columns
- **Tokenized** quasi-identifiers(QIs) using hashlib
- Performed **K-anonymity** : Ensured each unique combination of QIs appears at least $k=5$ times
- Performed **L-diversity** : Within each QI group , the **sensitive attribute**(Orientation) has **at least 1 different values**, preventing inference attacks
- This is done individually to each of the split dataset to simulate 3 different hospitals.

Interpretation :

- We anonymized the dataset using the above mentioned techniques to
 - Prevent any re-identification of an individual from the dataset.
 - Protect sensitive user data
 - Comply to privacy laws such as HIPPA, GDPR , DPDP.
 - Minimize Trust Requirements in Federated Learning

Algorithm for K-anonymity

For each record in dataset:

 Convert values in quasi-identifiers to strings Group dataset by (QI)

For each group:

 If group size $\geq k$:

 Keep all records in this group

Combine all valid groups to form k-anonymous dataset

Algorithm for L- diversity

Group the k-anonymous dataset by QIs again

For each group:

 If number of unique values in 'orientation' $\geq l$:

 Keep this group

Combine all valid groups to form l-diverse dataset

Module 3: Local Model Initialization

Tools used : PyTorch , sklearn , SMOTE , Opacus , TenSEAL

Process

1. Local Model Training with Differential Privacy

Preprocessing:

- Encode categorical features using Label Encoding.
- Apply SMOTE to balance classes.
- Split data into train/validation sets.
- Convert to PyTorch TensorDataset and DataLoader.

Initialize Model & Privacy:

- Define MLP model 'M'
- Initialize optimizer (Adam)
- Initialize Opacus PrivacyEngine

```
privacy_engine = PrivacyEngine()  
model, optimizer, train_loader =  
privacy_engine.make_private(  
    module=model,  
    optimizer=optimizer,  
    data_loader=train_loader,  
    noise_multiplier=1.0,  
    max_grad_norm=1.0,)
```

Training Loop (DP-aware):

For each epoch $e = 1$ to E :

Set model to training mode.

For each batch (x_a, y_b) in *train_loader* :

- a. Compute predictions: $y_b = M(x_b)$
- b. Compute loss: $L_b = \text{CrossEntropy}(y^{\wedge}_b, y_b)$
- c. Zero gradients: *optimizer.zero_grad()*
- d. Backpropagate: $L_b.\text{backward}()$
- e. **Opacus internally:**
 - Clips per-sample gradients to norm C
 - Adds Gaussian noise scaled by σ
- f. Update model: *optimizer.step()*

Evaluate on Validation Set:

- Disable gradient computation (`torch.no_grad()`)
- Compute predictions and validation loss.
- Store best model if validation improves.

Interpretation

1. We perform **local model training** to ensure that each hospital trains its model on **its own sensitive patient data**, without sharing raw data with any central server.
 - This preserves data locality and respects privacy regulations like HIPAA or GDPR.
2. To further strengthen privacy, we apply **differential privacy (DP)** during training.
 - DP adds carefully calibrated noise to gradients, ensuring that **no single patient's data has a significant impact** on the model's parameters.
 - This protects from Gradient Leakage, Membership Inference Attacks.

Module 4: Encryption of Gradients

Tools used : TenSEAL , torch , pickle

Process

1. Flatten gradients:

- **Standardization:** Flattening converts gradients of different shapes into a single 1D vector, making them uniform for processing.
- **Encryption Compatibility:** CKKS encryption requires inputs as flat vectors of floats
- **Communication & Aggregation:** Flattened gradients are easier to chunk, transmit securely, and aggregate on the server.

2. Chunk gradients if too long:

```
chunks = split(flattened_grads, chunk_size)
```

3. For each chunk:

- Convert to list
- Encrypt using CKKS:
 - `encrypted_chunk = ts.ckks_vector(context, chunk)`
- Serialize encrypted_chunk
- Append to encrypted_chunks list

4. Return encrypted_chunks

Interpretation

Homomorphic Encryption (HE) in federated learning (FL) pipeline plays a **critical role in preserving privacy during communication and aggregation.**

- **Secure Gradient Sharing:**
After local training, gradients are encrypted using HE to prevent the server from accessing raw, sensitive information.
- **Encrypted Aggregation at Server:**
The server aggregates encrypted gradients directly (via CKKS) without decrypting them, maintaining privacy even if the server is semi-honest.
- **Client-Side Decryption & Update:**
Only clients hold the decryption key, so only they can decrypt the aggregated global gradient and update their model securely.
- Protects against attacks such as Gradient Inversion of intermediate gradients , Membership Inference , Eavesdropping , Man-in-the-Middle Attacks.

Module 4: Secure Gradient Communication & Aggregation

Tools used : Flask , ssl , pickle , requests , shutil ,

Secure Communication

Ensures only the encrypted content reaches the server

- **Input:** Encrypted gradient chunks, Server URL
 - **Output:** Chunk files stored securely on the server
-
1. **For each gradient chunk:**
 - Serialize the CKKS vector using TenSEAL.
 - Wrap *chunk_id*, *client_id*, and data into a dictionary.
 - Serialize the dictionary using Pickle.
 - Send the payload via HTTPS POST to the Flask server.
 2. **Server receives the POST request:**
 - Deserialize the payload.
- Extract *chunk_id*, *client_id*, and encrypted gradient.
 - Save it in '*received_chunks_bin/{client_id}/chunk_{chunk_id}.bin*'

Aggregation

To aggregate encrypted gradient chunks from all clients without ever decrypting them, ensuring privacy even from the server.

Input: Encrypted gradient chunks from multiple clients

Output: Aggregated encrypted global gradient

1. Load CKKS context from '**context.ser**'.
2. For each client directory in '**received_chunks_bin**':
 - Load all '**.bin**' chunk files.
 - Deserialize and parse them into CKKS vectors.
 - Group them by chunk index.
3. For each chunk index:
 - Homomorphically add all CKKS vectors in the group.
 - Divide the result by number of clients (encrypted average).
4. Save:
 - Each aggregated encrypted chunk → '**aggregated_chunks_bin/**'.
 - All serialized chunks into → '**aggregated_gradient_global_encrypted.pkl**'.
5. Delete the '**received_chunks_bin**' folder for cleanup.

Module 5: Decryption and Global Module Update

Tools used : TenSEAL , pickle , Pytorch

Process

1. Receive Encrypted Gradient

- Client downloads the aggregated global gradient (aggregated_gradient_global_encrypted.pkl) from the server.

2. Load Private CKKS Context

- Client loads its TenSEAL CKKS context (with private key) to enable decryption.

3. Decrypt Gradient Chunks

- Each encrypted chunk is deserialized and decrypted using the CKKS vector to get floating-point gradient values.

4. Reconstruct Full Gradient

- Decrypted chunks are concatenated into a single flattened global gradient vector.

5. Clip Gradient Norm

- If the gradient norm is too large, it is scaled down to maintain numerical stability.

6. Update Model Weights

- The flattened gradient is split and reshaped to match model parameter sizes

7. Model Synchronize

- Client's model is now updated with global knowledge and ready for the next FL round or evaluation.

Project Demonstration

Exhibit the demonstration of the complete project.

- It has to be complete in all aspects
- Exhibit working of the project

Test Plan and Strategy

Test Methods

Functional Testing:

- Unit Testing: Validated modules like anonymization, encryption, and local training individual to ensure that they work correctly in isolation.
- Integration Testing: Ensured correct interaction between clients and server over SSL (Federated Learning) to make sure that they work smoothly.
- Federated Rounds Testing: Checked accuracy progression and consistency across multiple communication rounds.

Non-Functional Testing:

- Security Testing: Confirmed the effectiveness of Anonymization, DP and HE in protecting data, and validated defense against attacks like homogeneity attack, MITM etc.
- Performance Testing: Compared model accuracy with and without privacy techniques to assess computational and learning overhead.
- Robustness Testing: Simulated Byzantine & MITM attacks to observe system behavior under malicious conditions.

Test Plan and Strategy

Test Environment

Components	Details
Operating System	Windows 11 and macOS Sequoia.
Execution Tools	Jupyter Notebooks (clients), Flask (SSL server)
Dataset Used	Depression survey dataset (DASS)
Testing Setup	3 clients (simulated hospitals), 1 global server, Federated learning rounds until convergence.
Privacy Techniques used	Anonymization, DP, Homomorphic Encryption
Communication Setup	Secure communication via HTTPS (SSL/TLS), gradient chunking with pickle

Test Plan and Strategy

Team Roles

Team Member Responsible	Tasks
Diya	Implemented Federated Learning and Homomorphic Encryption, performed performance testing, and simulated security attacks (Byzantine, MITM).
Disha	Focused on anonymization-related attacks (Homogeneity, Gradient Leakage, Re-identification, Background Knowledge), performed data analysis and research documentation.
Disha & Diya	Collaboratively implemented Anonymization + Differential Privacy module and prepared result visualizations.

Test Plan and Strategy

Benefits of this approach

- **Modular Assurance:** Every component is independently verified (DP, HE, MLP, comms).
- **Privacy Guarantee:** Confirms encryption and DP prevent gradient/data leakage.
- **Scalable Testing:** Simulated clients help test larger deployments.
- **Early Bug Detection:** Unit/integration testing reduces runtime errors later.

Drawbacks

- **Performance Overhead:** Testing encryption-heavy systems is slower.
- **Limited Real-World Testing:** Simulated environment may not match real hospital network conditions.
- **Complex Debugging:** Encrypted data is hard to inspect, which complicates test failures.

Test Plan and Strategy

Test Tools Used

- **Open-Source Libraries:** PyTorch, Opacus, TenSEAL, Scikit-learn for model training, differential privacy, encryption, and evaluation.
- **Communication & Integration:** Flask was used to simulate secure client-server interactions in the federated learning setup.
- **Visualization & Debugging:** Matplotlib, Seaborn for performance graphs and trend analysis.
- **Manual Functional Testing:** Each module (anonymization, encryption, local model training, secure transmission) was tested via controlled scripts and monitored outputs.
- **No Automated Test Suites Used:** Since the project was research-oriented, emphasis was on accuracy, privacy, and robustness, not formal automation.
- **Development & Execution Environment:** Jupyter Notebook, Colab, VS Code

Results and Discussion

Anonymization

education	urban	gender	engnat	screensize	uniquenet	hand	religion	orientation	race	voted	married	familysize	age	Q3A	Q5A	Q10A	Q13A	Q16A	Q17A
TKN_25ff2	TKN_2915	Female	Yes	TKN_ea75	TKN_bdb4	TKN_92b0	TKN_05b8	TKN_1b57	TKN_25a8	TKN_93cb	TKN_f5c2c	3	Adult	3	4	3	4	4	4
TKN_25ff2	TKN_2915	Female	Yes	TKN_ea75	TKN_9134	TKN_92b0	TKN_05b8	TKN_1b57	TKN_25a8	TKN_93cb	TKN_f5c2c	3	Adult	1	1	3	2	2	2
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_6b76	TKN_25a8	TKN_bafd	TKN_3954	3	Adult	2	2	2	2	2	2
TKN_25ff2	TKN_2915	Female	Yes	TKN_ea75	TKN_9134	TKN_92b0	TKN_05b8	TKN_6b76	TKN_25a8	TKN_93cb	TKN_f5c2c	3	Adult	1	3	2	4	2	2
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_6b76	TKN_25a8	TKN_bafd	TKN_3954	3	Adult	1	2	1	2	1	2
TKN_25ff2	TKN_2915	Female	No	TKN_f1f2k	TKN_9134	TKN_945d	TKN_05b8	TKN_1b57	TKN_25a8	TKN_93cb	TKN_f5c2c	2	Adult	2	2	2	4	3	3
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_2c94	TKN_25a8	TKN_93cb	TKN_f5c2c	6	Adult	2	3	2	4	3	4
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_1b57	TKN_25a8	TKN_bafd	TKN_f5c2c	1	Adult	2	2	2	3	2	1
TKN_25ff2	TKN_2915	Female	Yes	TKN_ea75	TKN_9134	TKN_945d	TKN_05b8	TKN_1b57	TKN_25a8	TKN_93cb	TKN_f5c2c	3	Adult	2	4	2	4	3	4
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_1b57	TKN_25a8	TKN_93cb	TKN_f5c2c	2	Adult	2	2	2	3	2	3
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_1b57	TKN_25a8	TKN_93cb	TKN_f5c2c	1	Adult	2	2	2	2	2	4
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_1b57	TKN_25a8	TKN_bafd	TKN_f5c2c	3	Adult	3	3	2	4	3	2
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_6b76	TKN_25a8	TKN_bafd	TKN_3954	2	Adult	2	4	3	3	3	4
TKN_25ff2	TKN_2915	Female	Yes	TKN_ea75	TKN_9134	TKN_92b0	TKN_05b8	TKN_6b76	TKN_25a8	TKN_bafd	TKN_f5c2c	3	Adult	2	3	2	4	2	3
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_6b76	TKN_25a8	TKN_93cb	TKN_c769	2	Adult	3	3	3	4	3	3
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_945d	TKN_05b8	TKN_1b57	TKN_25a8	TKN_93cb	TKN_c769	4	Adult	3	4	4	4	4	4
TKN_25ff2	TKN_2915	Female	No	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_6b76	TKN_25a8	TKN_bafd	TKN_f5c2c	2	Adult	4	4	4	4	4	4
TKN_25ff2	TKN_2915	Female	Yes	TKN_ea75	TKN_9134	TKN_92b0	TKN_05b8	TKN_6b76	TKN_25a8	TKN_93cb	TKN_f5c2c	2	Adult	1	3	2	2	2	2
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_1b57	TKN_25a8	TKN_bafd	TKN_3954	2	Adult	3	2	4	4	3	4
TKN_25ff2	TKN_2915	Female	Yes	TKN_ea75	TKN_9134	TKN_92b0	TKN_05b8	TKN_6b76	TKN_25a8	TKN_bafd	TKN_3954	3	Adult	3	4	4	4	4	4
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_6b76	TKN_25a8	TKN_bafd	TKN_f5c2c	2	Adult	2	1	2	2	1	1
TKN_25ff2	TKN_2915	Female	No	TKN_ea75	TKN_9134	TKN_92b0	TKN_05b8	TKN_6b76	TKN_25a8	TKN_93cb	TKN_f5c2c	2	Adult	2	4	2	2	2	3
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_4df4	TKN_25a8	TKN_93cb	TKN_f5c2c	2	Adult	4	4	4	4	4	4
TKN_25ff2	TKN_2915	Female	Yes	TKN_f1f2k	TKN_9134	TKN_92b0	TKN_05b8	TKN_1b57	TKN_25a8	TKN_bafd	TKN_f5c2c	2	Adult	4	4	4	4	4	4
TKN_25ff2	TKN_2915	Female	Yes	TKN_ea75	TKN_9134	TKN_92b0	TKN_05b8	TKN_6b76	TKN_25a8	TKN_bafd	TKN_f5c2c	2	Adult	2	4	3	2	4	2

Results and Discussion

Initial Local Model Accuracy for the main framework (FL with Anonymization + DP + HE)

Training Accuracy: 0.7812
Validation Accuracy: 0.7635

Classification Report (Validation):

	precision	recall	f1-score	support
Extremely Severe	0.91	0.90	0.91	794
Mild	0.44	0.74	0.55	220
Moderate	0.73	0.64	0.68	432
Normal	0.98	0.72	0.83	511
Severe	0.61	0.69	0.65	377
accuracy			0.76	2334
macro avg	0.73	0.74	0.72	2334
weighted avg	0.80	0.76	0.77	2334

Differential Privacy Epsilon: 1.86 ($\delta=1e-5$)

Client 1 Initial accuracy

Results and Discussion

Final Accuracy Analysis for the main framework (FL with Anonymization + DP + HE)

```
[CLIENT] Training → Accuracy: 83.58%, Loss: 95.7938
[CLIENT] Validation → Accuracy: 82.26%, Loss: 14.4990
[CLIENT] Validation Classification Report:
      precision    recall  f1-score   support

     0       0.94      0.91      0.93       794
     1       0.56      0.81      0.67       220
     2       0.84      0.63      0.72       432
     3       0.98      0.85      0.91       511
     4       0.65      0.83      0.73       377

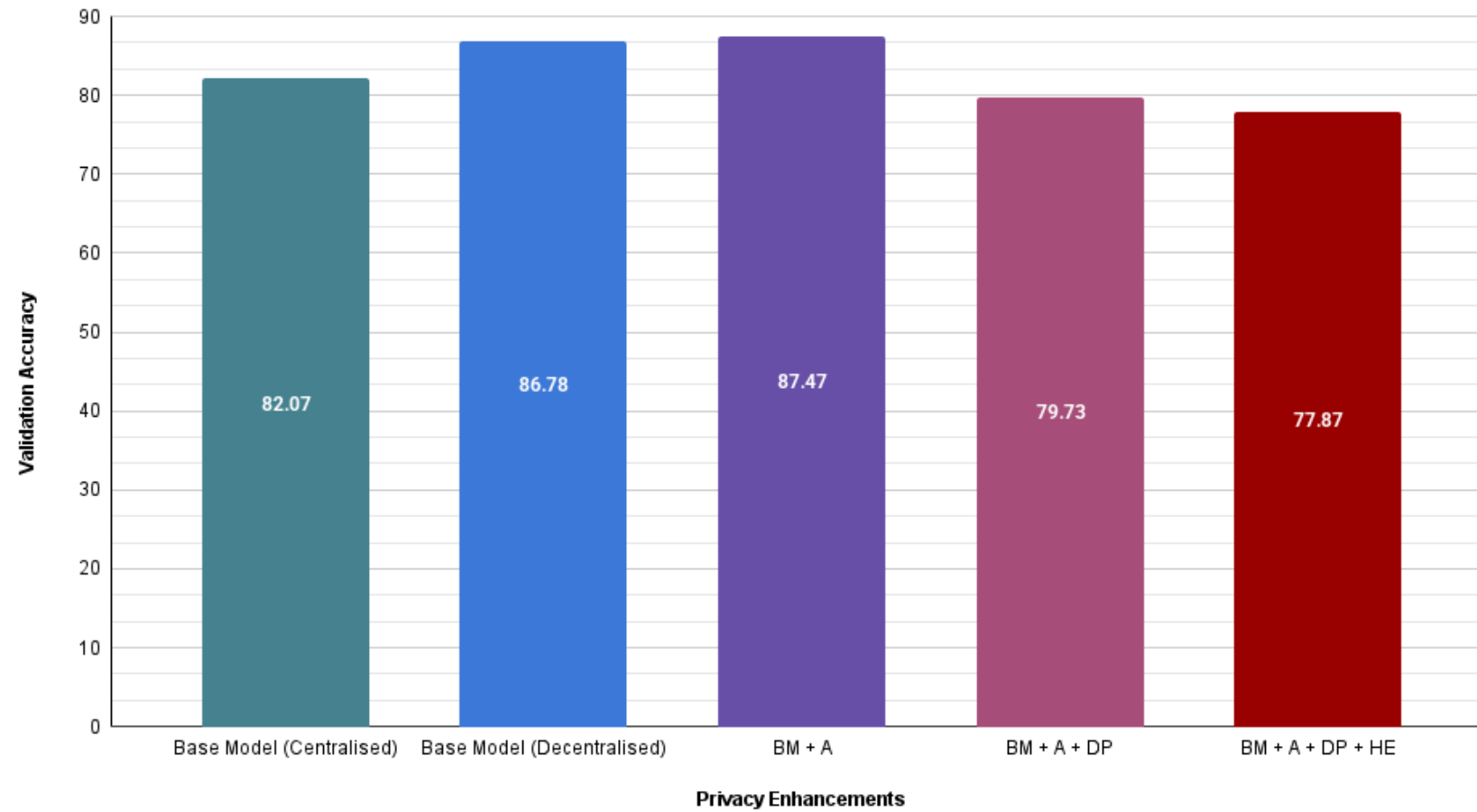
 accuracy          0.82       2334
 macro avg         0.79      0.81      0.79       2334
 weighted avg      0.85      0.82      0.83       2334

Best model saved.

|== Federated Round 11 ==|
```

Client 1 final accuracy

Local Model Evaluation Metrics



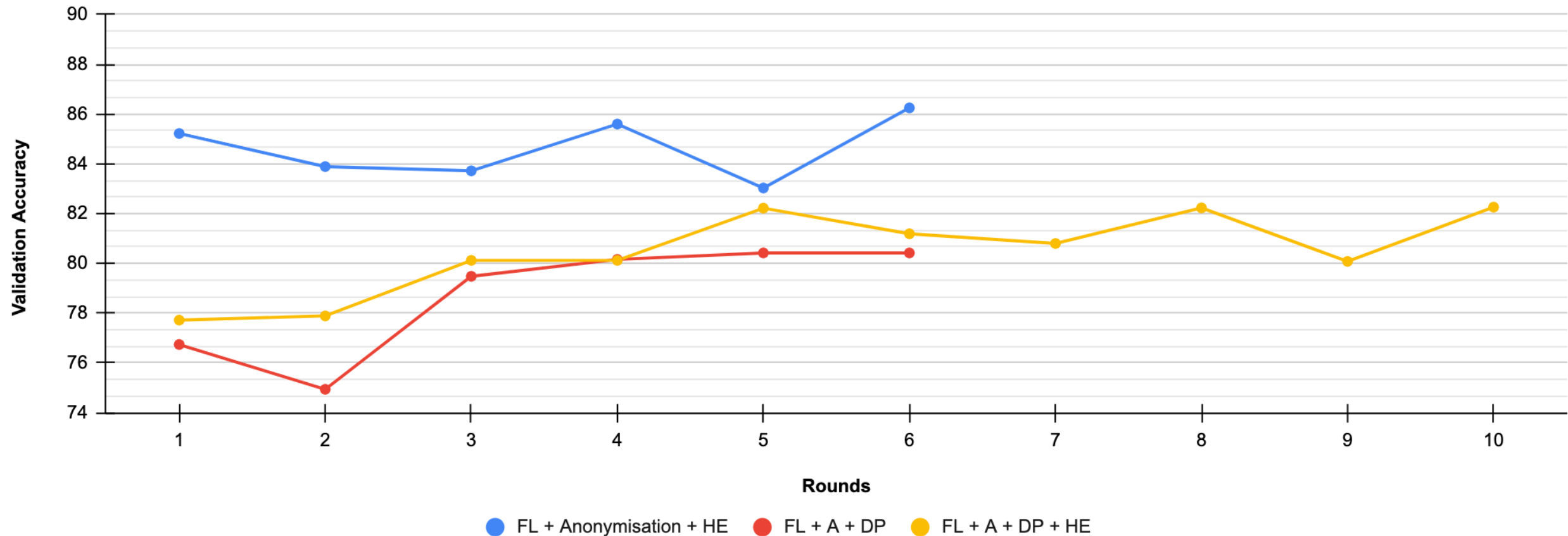
Results and Discussion

Accuracy Analysis across multiple setups

- **Main Framework (FL + Anonymization + DP + HE):** Final global model accuracy across hospitals: **78% - 82%**
- **FL with HE + Anonymization (No DP):** Achieved slightly higher global accuracy: ~ **80% - 89%**. This indicates privacy-utility trade-off due to DP noise.
- **FL with Anonymization + DP (No HE):** Final global validation accuracy remained consistent (~**78% – 82%**), showing that Homomorphic Encryption has minimal impact on model performance, and core privacy is preserved effectively through anonymization + DP alone.
- **Local training with anonymization** improved client performance: Hospital 1 jumped from **86.78% to 87.47%** accuracy after anonymization. Anonymization likely acted as regularization, reducing overfitting and increasing homogeneity.

Evaluation Metrics

Federated Learning Evaluation



Evaluation Metrics - ARX (Risk Analysis)



Overview		Population uniques
Measure	Value [%]	
Lowest prosecutor risk	0.10299%	
Records affected by lowest risk	8.62881%	
Average prosecutor risk	2.9681%	
Highest prosecutor risk	20%	
Records affected by highest risk	1.19968%	
Estimated prosecutor risk	20%	
Estimated journalist risk	20%	
Estimated marketer risk	2.9681%	
Sample uniques	0%	
Population uniques	0%	
Population model	DANKAR	

Attack Analysis

Linkage Attack

We simulate an attacker with access to auxiliary background knowledge derived from a subset of the raw data prior to anonymization.

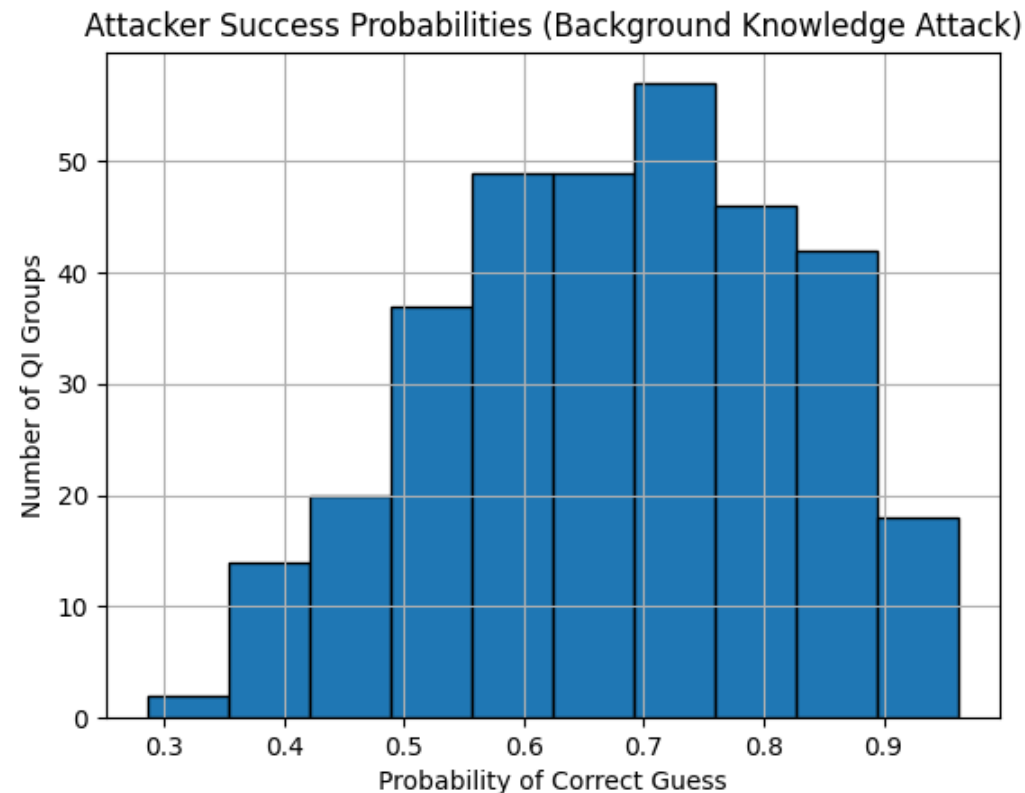
The attacker uses common quasi-identifiers (age, gender, religion) to perform linkage with the anonymized dataset and assess re-identification risk

```
Total auxiliary records: 12874  
Uniquely re-identified individuals (k=1): 100  
Re-identification rate: 0.78%
```

Background Knowledge Attack

Background knowledge attacks occur when an attacker uses external information—like age, gender, or race code—to match individuals in an anonymized dataset and reveal their sensitive data. These attacks exploit quasi-identifiers, making it possible to re-identify individuals even when names are removed.

- Average attacker success probability (background knowledge): 67.45%



Homogeneity Attack

A **homogeneity attack** is a privacy breach where an attacker can infer sensitive information about individuals **because all records in an anonymized group (QI group) share the same sensitive value**.

- Even if a dataset satisfies **k-anonymity** (e.g., 5 people share the same quasi-identifiers like age, gender, location), it can still leak sensitive information if **everyone in that group has the same sensitive attribute**.
- The attacker doesn't need to know which exact record belongs to the victim—because **all records say the same thing**, so the sensitive value is exposed anyway.

Result

With k anonymity and l diversity

Total QI groups: 334

Groups vulnerable to homogeneity attack: 0

Proportion of leaking groups: 0.00%

Only k anonymity

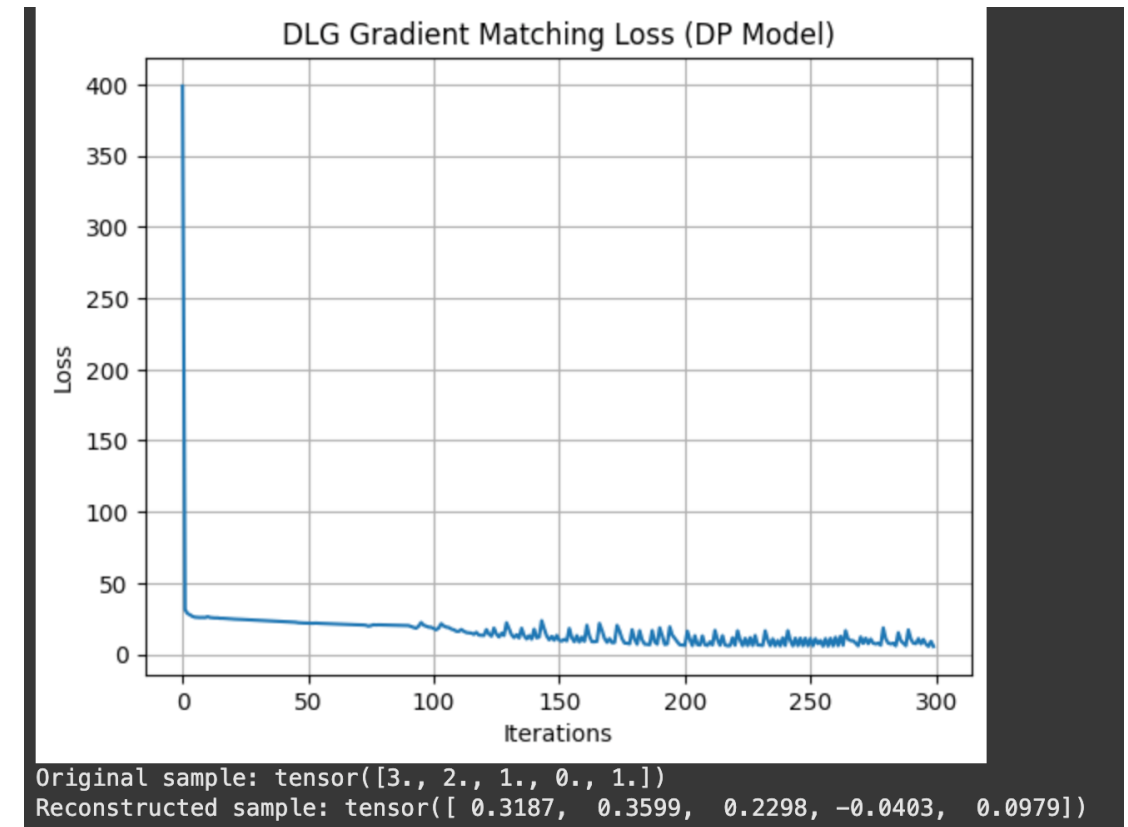
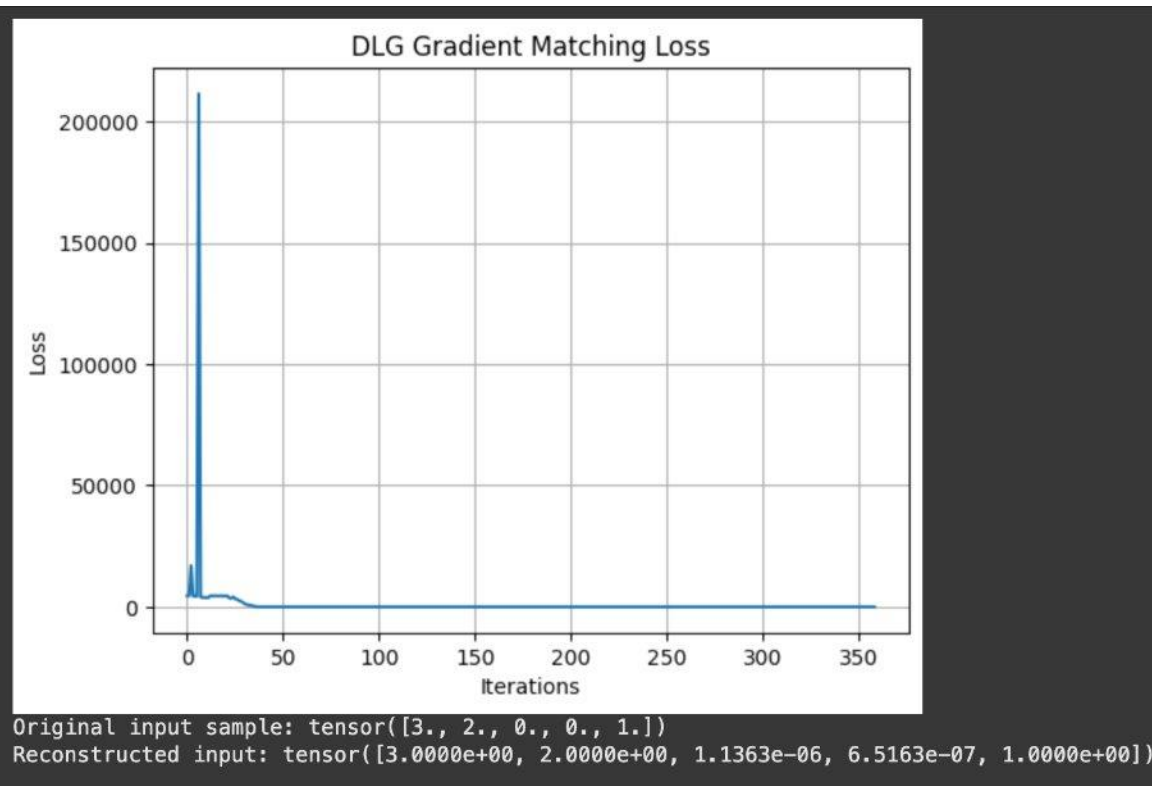
Total QI groups: 364

Groups vulnerable to homogeneity attack: 30

Proportion of leaking groups: 8.24%

Gradient Leakage Attack (without Homomorphic Encryption)

A **gradient leakage attack** allows an adversary to reconstruct raw training data from shared gradients in federated learning, compromising user privacy unless proper defenses like **DP** or **encryption** are used.



Gradient Leakage Attack (with Homomorphic Encryption)

Encrypted gradient for model.0.weight:

b'\n\x04\x80 \x80\x06\x12\x9c\xb1\x14^\xa1\x10\x04\x01\x02\x00\x00\x9c\x18\x05\x00\x00\x00\x00(\xb5/\xfd\xa0a\x00\x06

Encrypted gradient for model.0.bias:

b'\n\x02\x80\x01\x12\xa1\xb5\x14^\xa1\x10\x04\x01\x02\x00\x00\xa1\x1a\x05\x00\x00\x00\x00(\xb5/\xfd\xa0a\x00\x06\x00\x

Encrypted gradient for model.2.weight:

b"\n\x04\x80 \x80 \x12\xa8\xb2\x14^\xa1\x10\x04\x01\x02\x00\x00(\x19\x05\x00\x00\x00\x00(\xb5/\xfd\xa0a\x00\x06\x00\x

Encrypted gradient for model.2.bias:

b'\n\x01@\x12\xee\xb4\x14^\xa1\x10\x04\x01\x02\x00\x00n\x1a\x05\x00\x00\x00\x00(\xb5/\xfd\xa0a\x00\x06\x00\xbcZ\x0e\x

Encrypted gradient for model.4.weight:

b'\n\x02\xc0\x02\x12\xd1\xb2\x14^\xa1\x10\x04\x01\x02\x00\x00Q\x19\x05\x00\x00\x00\x00(\xb5/\xfd\xa0a\x00\x06\x00\x14

Encrypted gradient for model.4.bias:

b"\n\x01\x05\x12\xc9\xb3\x14^\xa1\x10\x04\x01\x02\x00\x00\xc9\x19\x05\x00\x00\x00\x00(\xb5/\xfd\xa0a\x00\x06\x00\x0c [

Results and Discussion

Attack Analysis

- **Byzantine Attack:** When gradients were scaled (5000 initially, then 100),
 - Without gradient clipping, validation accuracy dropped drastically to **~33–34%**.
 - With gradient clipping (as in our privacy framework), accuracy remained stable at **~81–82%**, proving the robustness of our gradient protection.

```
[CLIENT] Training → Accuracy: 20.05%, Loss: nan
[CLIENT] Validation → Accuracy: 33.42%, Loss: nan
[CLIENT] Validation Classification Report:
      precision    recall  f1-score   support

     0       0.33       1.00       0.50        782
     1       0.00       0.00       0.00        234
     2       0.00       0.00       0.00        411
     3       0.00       0.00       0.00        517
     4       0.00       0.00       0.00        396

 accuracy                   0.33        2340
 macro avg       0.07       0.20       0.10        2340
 weighted avg    0.11       0.33       0.17        2340

Early stopping.
```

Without Gradient Clipping

```
[CLIENT] Training → Accuracy: 83.34%, Loss: 96.9314
[CLIENT] Validation → Accuracy: 82.31%, Loss: 14.5145
[CLIENT] Validation Classification Report:
      precision    recall  f1-score   support

     0       0.95       0.89       0.92        782
     1       0.58       0.76       0.66        234
     2       0.77       0.77       0.77        411
     3       0.99       0.82       0.89        517
     4       0.68       0.79       0.73        396

 accuracy                   0.82        2340
 macro avg       0.79       0.81       0.79        2340
 weighted avg    0.84       0.82       0.83        2340C
```

With Gradient Clipping

Results and Discussion

Attack Analysis

Man-in-the-Middle (MITM) Attack:

- With HE enabled, tampering failed as the gradients were encrypted, and only metadata like timestamp and payload size was visible.
- Without HE, unencrypted gradients from Client 1 were intercepted and tampered mid-transit via a proxy server (scaling done after gradient clipping).
- **Result:** Client 1's validation accuracy was ~79%, while other clients dropped to ~10–18%, showing severe model drift due to compromised updates.

Results and Discussion

Man In The Middle (MITM) Attack Analysis

```
[MITM] MITM Proxy listening on http://127.0.0.1:9090 ...
* Serving Flask app '__main__'
* Debug mode: off
```

```
[MITM] Intercepted chunk 0 from hospital_1
Time: 2025-07-22 08:44:28
[MITM] Unknown bytes format
[MITM] Could not tamper – data is encrypted or not float array.
[MITM] Forwarded chunk 0 to real server.
```

```
[MITM] Total chunks from hospital_1: 2
```

```
[MITM] Intercepted chunk 1 from hospital_1
Time: 2025-07-22 08:44:28
[MITM] Unknown bytes format
[MITM] Could not tamper – data is encrypted or not float array.
```

```
[MITM] Forwarded chunk 1 to real server.
```

```
[MITM] Total chunks from hospital_1: 2
```

With HE

```
[CLIENT] Training → Accuracy: 19.67%, Loss: 2409.8584
[CLIENT] Validation → Accuracy: 10.21%, Loss: 1188.6085
[CLIENT] Validation Classification Report:
```

	precision	recall	f1-score	support
0	0.17	0.00	0.00	782
1	0.10	0.99	0.18	234
2	0.00	0.00	0.00	411
3	0.75	0.01	0.02	517
4	0.00	0.00	0.00	396
accuracy			0.10	2340
macro avg	0.20	0.20	0.04	2340
weighted avg	0.23	0.10	0.02	2340

```
=== Federated Round 6 ===
```

Without HE

Timeline - Update on Tasks

JUNE	WEEK 1	WEEK 2	WEEK 3	WEEK 4
Literature Survey 1. Look into relevant research papers 2. Formulate problem statement 3. Create a work plan 4. Make a proposed work flow diagram	Look into relevant research papers (2/6/25 - 13/6/25) Formulate problem statement (4/6/25) Create a work plan (5/6/25) Make a proposed work flow diagram (5/6/25)			
Module 1 1. Raw data Collection 2. Cleaning of Dataset 3. Anonymization of dataset 4. Referred to papers for implementation of the algorithm and code 5. Learn ML basics	Raw data Collection (6/6/25)	Cleaning of Dataset (9/6/25) Anonymization of dataset (10/6/25 - 12/6/25) Referred to papers for implementation of the algorithm and code (11/6/25) Learn ML basics (13/6/25)		
Module 2 1. Feature Extraction 2. Local model initialisation 3. Add differential privacy while training the local model			Feature Extraction (16/6/25 - 17/6/25) Local Model Initialisation (16/6/25 - 18/6/25)	Add DP to model training (23/6/25)

Timeline - Update on Tasks

JULY	WEEK 1	WEEK 2	WEEK 3	WEEK 4
Module 3 1. Applying Homomorphic Encryption to the gradients of local models 2. Encryption using public key 3. Communication protocol for client-server exchange of encrypted gradients	Applying Homomorphic Encryption to the gradients of local models Encryption using public key Communication Protocol			
Module 4 1. Secure aggregation 2. Federated learning implementation 3. Anonymization Attacks 4. Man In The Middle 5. Byzantine Attack		Secure Aggregation Federated learning implementation Anonymization Attacks Man In The Middle Byzantine Attack		
Research paper 1. Final Review Preparation 2. Internship report				Final Review Preparation Internship Report

Conclusion and Future work

Conclusion:

Developed a privacy-preserving Federated Learning (FL) framework using:

- **Anonymization** techniques (k-Anonymity, l-Diversity)
- **Differential Privacy (DP)** for noise-based protection
- **Homomorphic Encryption (HE)** for secure gradient communication
- Tested against real-world privacy threats: Byzantine attacks, MITM interception, Gradient Leakage, Homogeneity, Background Knowledge, and Re-identification attacks
- Fine-tuned DP parameters for optimal trade-off between privacy and accuracy
- Applied HE-based aggregation without server-side decryption, ensuring true zero-trust server behavior

Conclusion and Future work

Future Works:

- **Integration of Federated Learning with Real-time Systems:** Extend the framework to operate with real-time medical records in hospital networks.
- **Support for Non-IID Data Distribution:** Extend the framework to handle heterogeneous data across clients, reflecting real-world scenarios where patient data distributions vary significantly.
- **Scalability to More Clients:** Optimize performance and communication to enable large-scale federated learning across dozens or hundreds of hospitals, ensuring efficiency and privacy are preserved.
- **Blockchain-based Federated Learning:** Integrate blockchain to ensure tamper-proof logging of model updates and guarantee transparency in client contributions.
- **Simulate further attacks** such as Side channel attack, Sybil , data poisoning attacks.

Anonymisation

“Anonymization Techniques for Privacy Preserving Data Publishing: A Comprehensive Survey | IEEE Journals & Magazine | IEEE Xplore.” Accessed: Jul. 24, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/9298747>

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N. Ponomareva *et al.*, “How to DP-fy ML: A Practical Guide to Machine Learning with Differential Privacy,” *jair*, vol. 77, pp. 1113–1201, Jul. 2023, doi: [10.1613/jair.1.14649](https://doi.org/10.1613/jair.1.14649).

Homomorphic Encryption

M. Gul, *Fully Homomorphic Encryption with Applications to Privacy-Preserving Machine Learning*, Bachelor's thesis, Harvard College, 2023.

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W. Jin *et al.*, “FedML-HE: An Efficient Homomorphic-Encryption-Based Privacy-Preserving Federated Learning System,” Jun. 17, 2024, *arXiv*: arXiv:2303.10837. doi: [10.48550/arXiv.2303.10837](https://doi.org/10.48550/arXiv.2303.10837).

R. Xu, N. Baracaldo, Y. Zhou, A. Anwar, and H. Ludwig, ‘HybridAlpha: An Efficient Approach for Privacy-Preserving Federated Learning’, in *Proceedings of the 12th ACM Workshop on Artificial Intelligence and Security*, London United Kingdom: ACM, Nov. 2019, pp. 13–23. doi: [10.1145/3338501.3357371](https://doi.org/10.1145/3338501.3357371).

R. Aziz, S. Banerjee, S. Bouzefrane, and T. Le Vinh, “Exploring homomorphic encryption and differential privacy techniques towards secure federated learning paradigm,” *Future Internet*, vol. 15, no. 9, p. 310, Sep. 2023. [Online]. Available: <https://doi.org/10.3390/fi15090310>

Attacks/Vulnerabilities

N. Bouacida and P. Mohapatra, “Vulnerabilities in Federated Learning,” *IEEE Access*, vol. 9, pp. 63229–63249, 2021, doi: [10.1109/ACCESS.2021.3075203](https://doi.org/10.1109/ACCESS.2021.3075203).

V. Mothukuri, R. M. Parizi, S. Pouriyeh, Y. Huang, A. Dehghantanha, and G. Srivastava, ‘A survey on security and privacy of federated learning’, *Future Generation Computer Systems*, vol. 115, pp. 619–640, Feb. 2021, doi: [10.1016/j.future.2020.10.007](https://doi.org/10.1016/j.future.2020.10.007).

IEEE Draft (Optional)

Display the draft of your project work in an IEEE template if you have prepared to submit it in a Conference/Journal.

Indicate the status of completion as well.

Download the IEEE template from the given link below,

<https://www.ieee.org/content/dam/ieee-org/ieee/web/org/conferences/Conference-template-A4.doc>

Thank You